RAJALAKSHMI ENGINEERING COLLEGE

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P2

Started on	Thursday, 24 September 2026, 11:16 AM
State	Finished
Completed on	Thursday, 24 September 2026, 11:21 AM
Time taken	4 mins 43 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int a[n];
    int dp[n];
    for(int i=0;i<n;i++){
        scanf("%d",&a[i]);
        dp[i]=1;
    }
    for(int i=1;i<n;i++){
        for(int j=0;j<i;j++){
            if(a[j]<=a[i]&& dp[j]+1>dp[i]){
                dp[i]=dp[j]+1;
            }
        }
    }
    int max=0;
    for(int i=0;i<n;i++){
        if(dp[i]>max){
            max=dp[i];
        }
    }
    printf("%d",max);
    return 0;
}
```

	Input	Expected	Got	
✔	9 -1 3 4 5 2 2 2 2 3	6	6	✔
✔	7 1 2 2 4 5 7 6	6	6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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